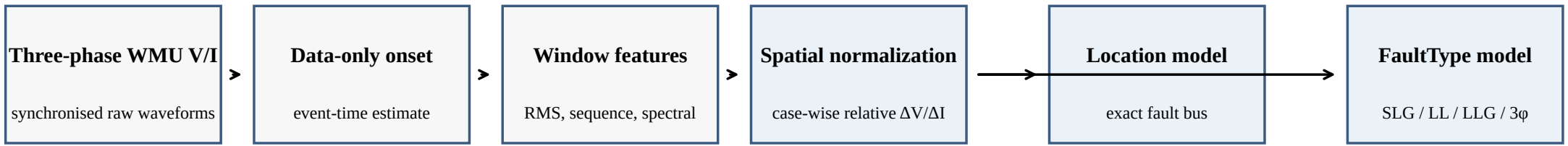
Three-phase WMU V/I
synchronised raw waveforms
Data-only onset
event-time estimate
Window features
RMS, sequence, spectral
Spatial normalization
case-wise relative ΔV/ΔI
Location model
exact fault bus
FaultType model
SLG / LL / LLG / 3φ
Separate supervised models
(no location/type conflation)